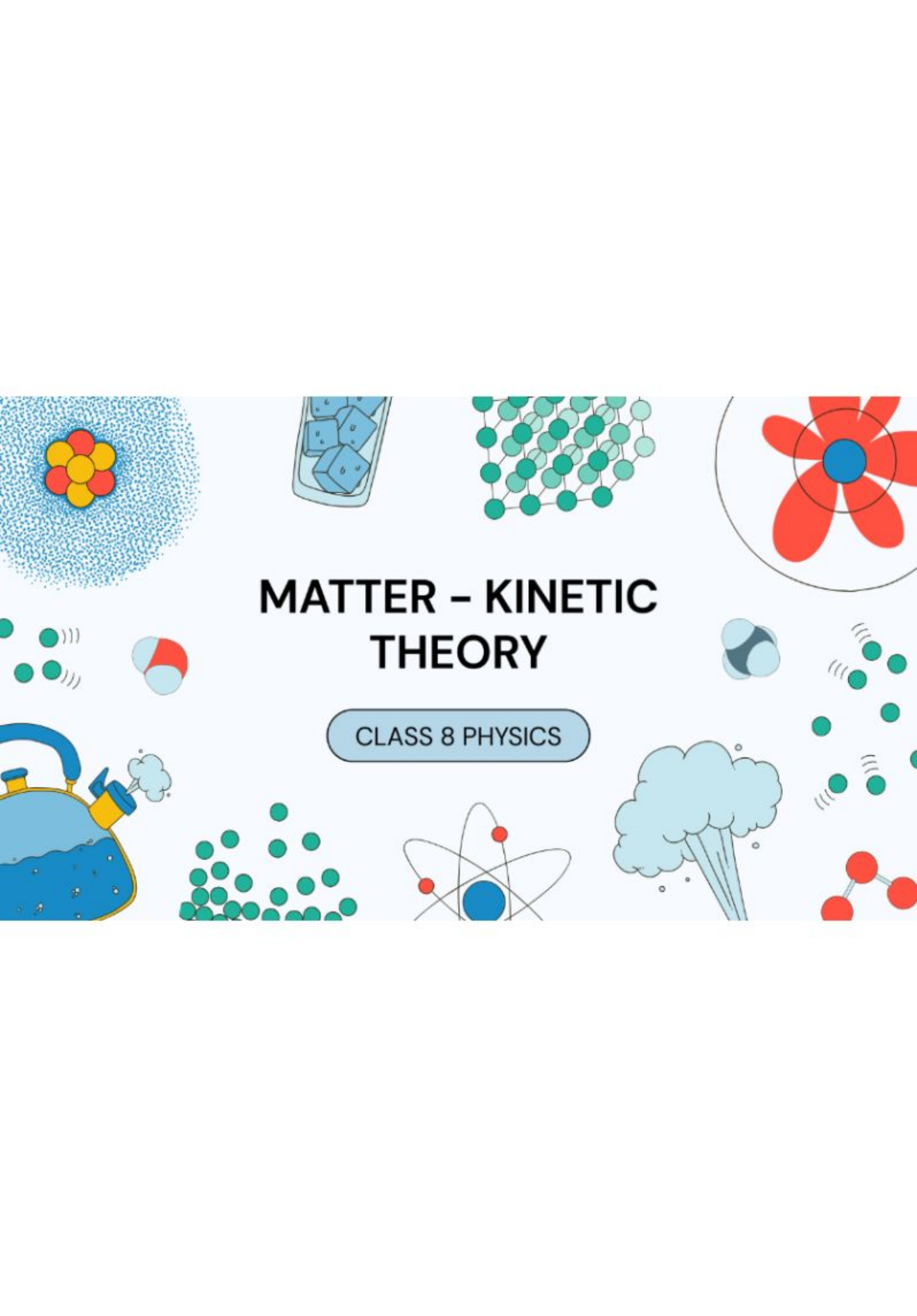


The page features a light blue background with a decorative border of various physics-related illustrations. At the top left is a cluster of yellow and red spheres. To its right is a tray of blue ice cubes. Further right is a 3D lattice of green spheres. At the top right is a red flower-like shape with a blue center. On the left side, there are green dots with motion lines and a red and white sphere. On the right side, there is a blue and white sphere, green dots with motion lines, and a red molecular structure. At the bottom left is a blue kettle with steam. At the bottom center is a Bohr-style atomic model with a blue nucleus and red electrons. At the bottom right is a blue cloud with rain. The title 'MATTER - KINETIC THEORY' is centered in the middle, and 'CLASS 8 PHYSICS' is in a blue box below it.

MATTER - KINETIC THEORY

CLASS 8 PHYSICS

Learning Goals

1. Three states of matter in terms of movement of particles.
2. Energy content in the three states of matter.
3. Change of state in matter using the Kinetic theory:
 - Boiling
 - Vaporization
 - Melting
 - Fusion
 - Evaporation
 - Condensation
 - Sublimation
 - Deposition
 - Freezing
4. Change of state diagrams (using the terms mentioned above).

What is the Kinetic Theory of Matter?

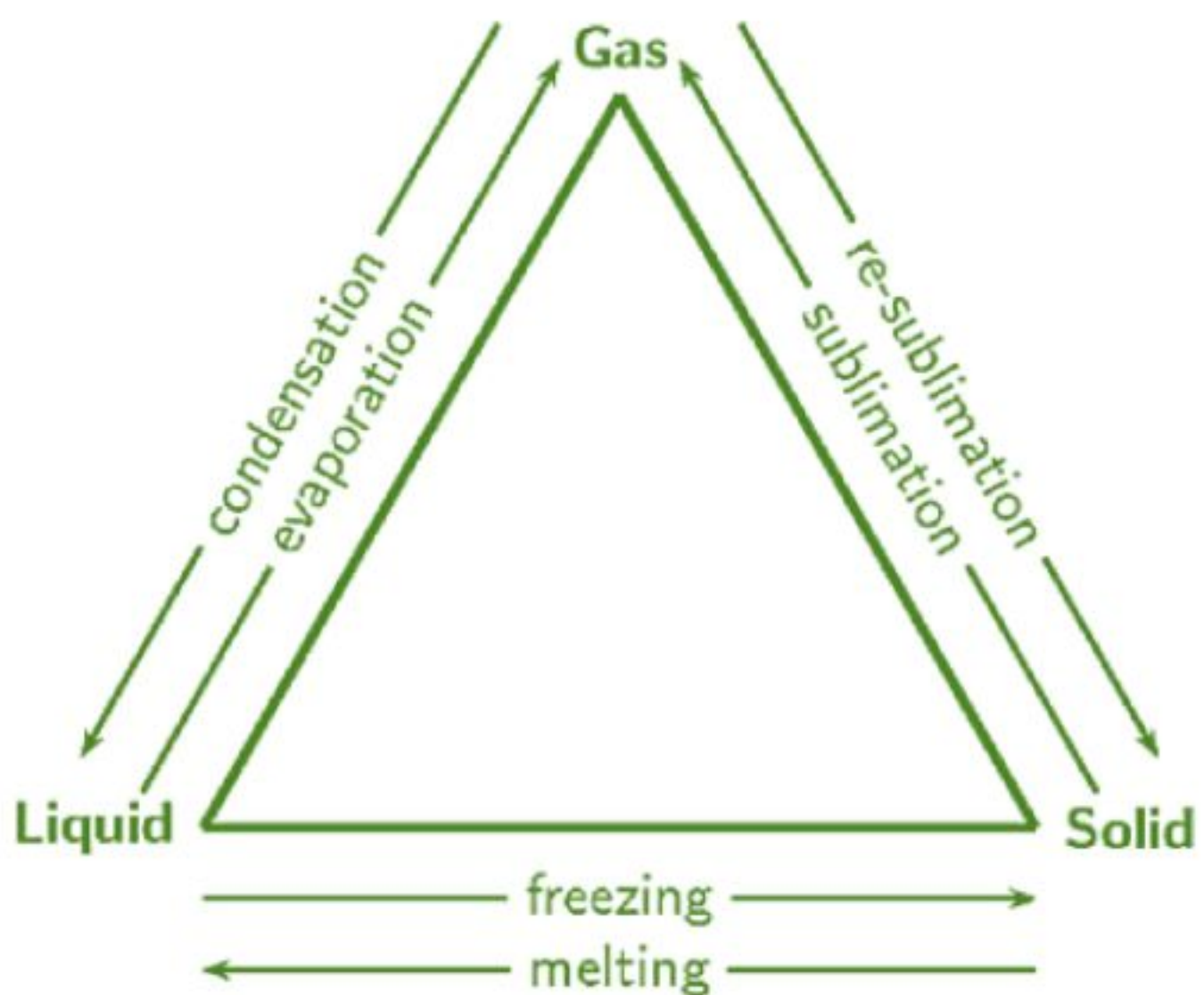
The Kinetic Theory of Matter states:

- All matter is made up of tiny particles (atoms, molecules, or ions).
-
- These particles are always in motion – they never stop moving.
-
- The motion of particles gives them kinetic energy.
-
- The temperature of a substance tells us how much average kinetic energy its particles have.
-
- Higher temperature → faster motion → more kinetic energy.
-
- Lower temperature → slower motion → less kinetic energy.
-
- Particles also experience forces of attraction between them (intermolecular forces).

Key idea: The state of matter (solid, liquid, gas) depends on:

How fast the particles are moving (kinetic energy).

How strongly they attract each other.



What is "Matter"?



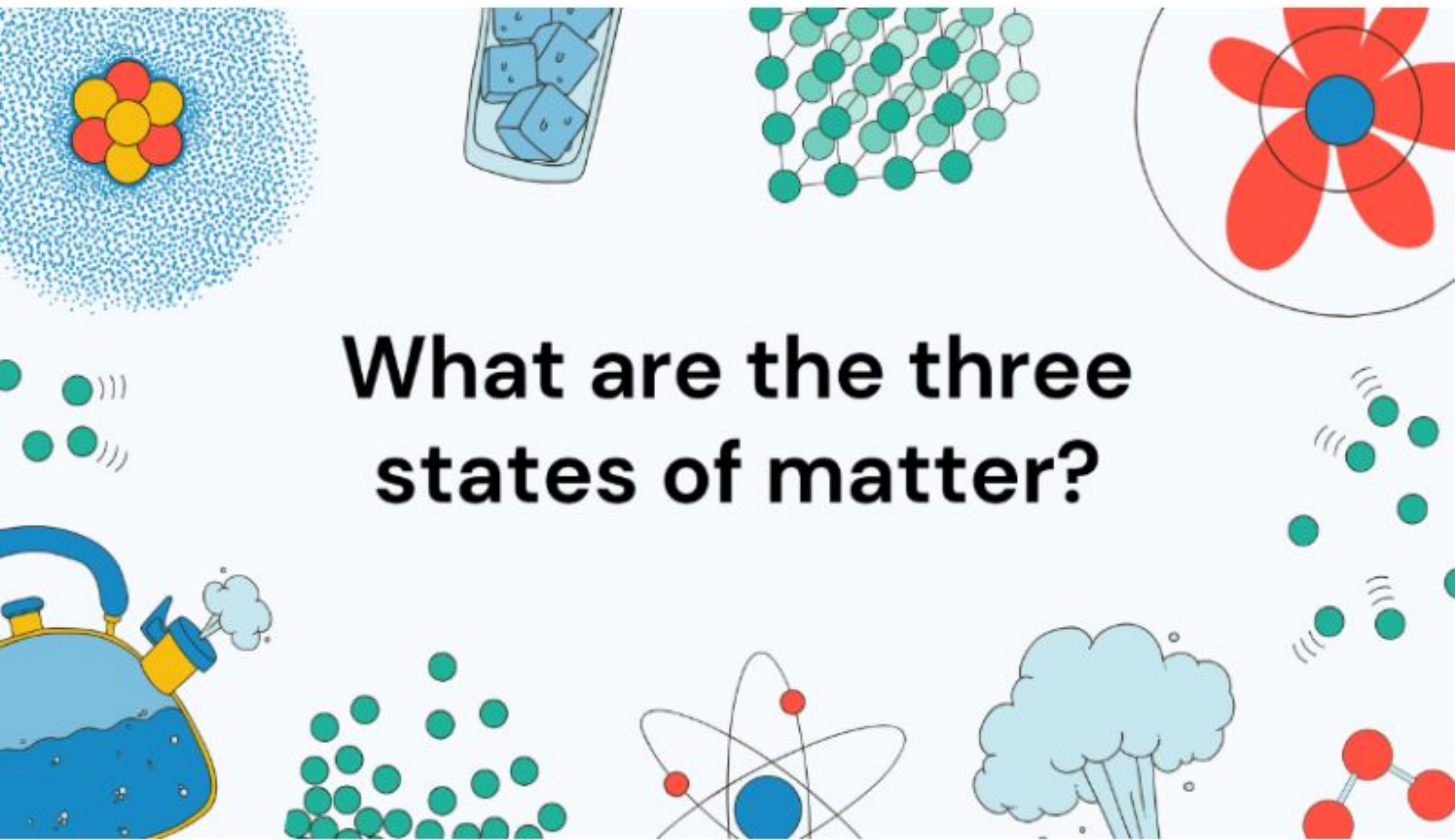
What is Matter?

Matter is any substance that has mass and takes up space (volume).

This means that anything you can touch, hold, or measure is matter.

For example, air, water, rocks, books and even your body are all made of matter!





**What are the three
states of matter?**

Main States of Matter

The three most common states of matter are:



Solid



Liquid



Gas

Properties of Matter

SOLID

- Arrangement: Particles are tightly packed in a regular pattern.
- Movement: Particles vibrate about fixed positions; they do not move from place to place.
- Intermolecular space: Very small.
- Intermolecular force: Very strong.
- Shape and volume: Fixed shape and fixed volume.
- Example: Ice, iron, wood.



Properties of Matter



LIQUID

- Arrangement: Particles are close together but not in a fixed pattern.
- Movement: Particles move around each other, sliding and flowing.
- Intermolecular space: More than solid, less than gas.
- Intermolecular force: Weaker than solid, stronger than gas.
- Shape and volume: Fixed volume, but no fixed shape (takes the shape of the container).
- Example: Water, oil, milk.



Properties of Matter

GAS

- Arrangement: Particles are far apart with no regular arrangement.
- Movement: Particles move rapidly and freely in all directions.
- Intermolecular space: Very large.
- Intermolecular force: Very weak (almost negligible).
- Shape and volume: No fixed shape and no fixed volume (expands to fill the container).
- Example: Water vapour, oxygen, air.

The Particle Model



Solid

Particles are tightly packed, strongly attract and vibrate in place.



Liquid

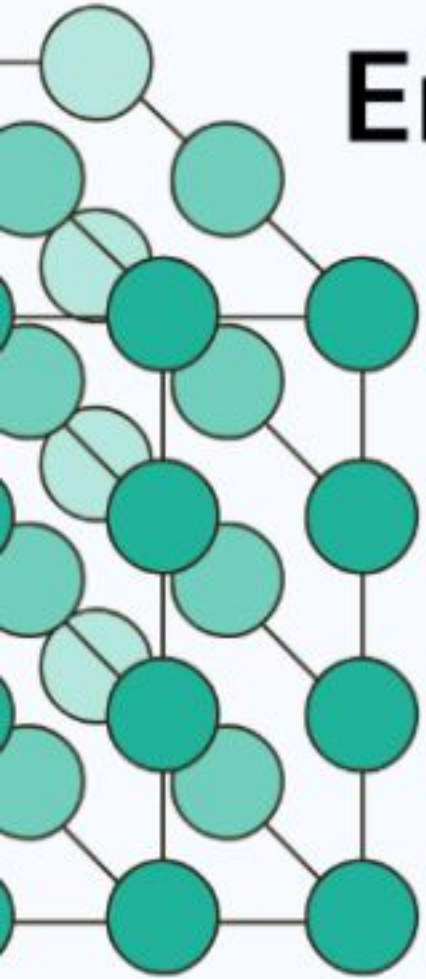
Particles are close together, attract and move past one another.



Gas

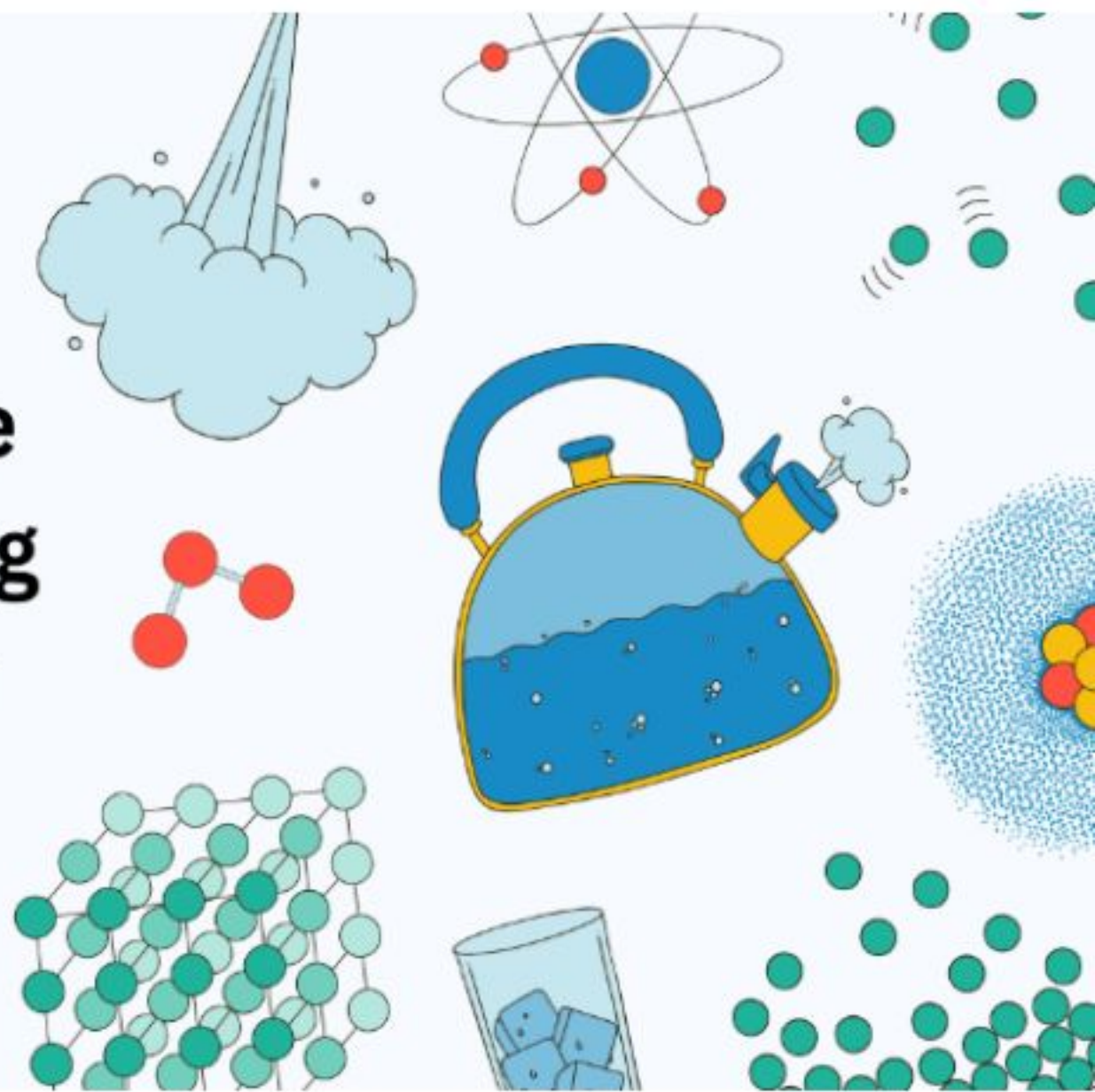
Particles are far apart with weak attraction and move freely.

Energy content in three states of matter

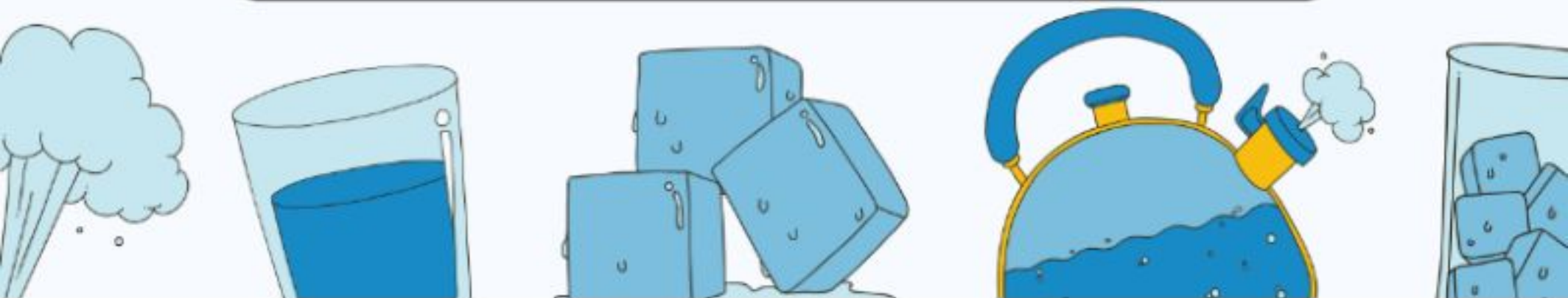


State	Average Kinetic Energy of Particles	Energy Content (relative)
Solid	Lowest (only vibration)	Lowest velocity=1
Liquid	Medium (vibration + sliding motion)	Medium velocity=1
Gas	Highest (fast, free motion)	Highest velocity=1

Change of State in Matter – Using Kinetic Theory



When we add heat to a substance, its particles gain kinetic energy and move more vigorously.
When we remove heat, particles lose kinetic energy and move more slowly.
This change in energy causes change of state.



(a) Melting (Fusion)

Occurs at the melting point of a substance.

Heat energy is absorbed.

Particles in the solid vibrate more vigorously.

At melting point, they overcome the forces holding them in fixed positions.

They start sliding past each other → solid becomes liquid.

Example: Ice → Water at 0°C.

Melting = Fusion (two names for the same process)

(b) Boiling

Occurs at the boiling point of a liquid.

Heat energy is absorbed.

Particles throughout the liquid gain enough energy to escape as gas.

Bubbles of gas form inside the liquid and rise to the surface.

Example: Water → Water vapour at 100°C (at sea level).

(c) Vaporization

Vaporization is the general term for liquid → gas change.

It includes:

Boiling (throughout the liquid, at boiling point)

Evaporation (only at surface, at any temperature).

(d) Condensation

Gas → Liquid.

Gas particles lose heat energy, slow down, and come closer.

Attractive forces pull them together to form liquid droplets.

(e) sublimation

Solid → Gas directly, without becoming liquid.

Particles gain enough energy to break free directly from the solid structure.

Examples:

Dry ice (solid CO_2) → carbon dioxide gas.

Naphthalene balls shrinking over time.

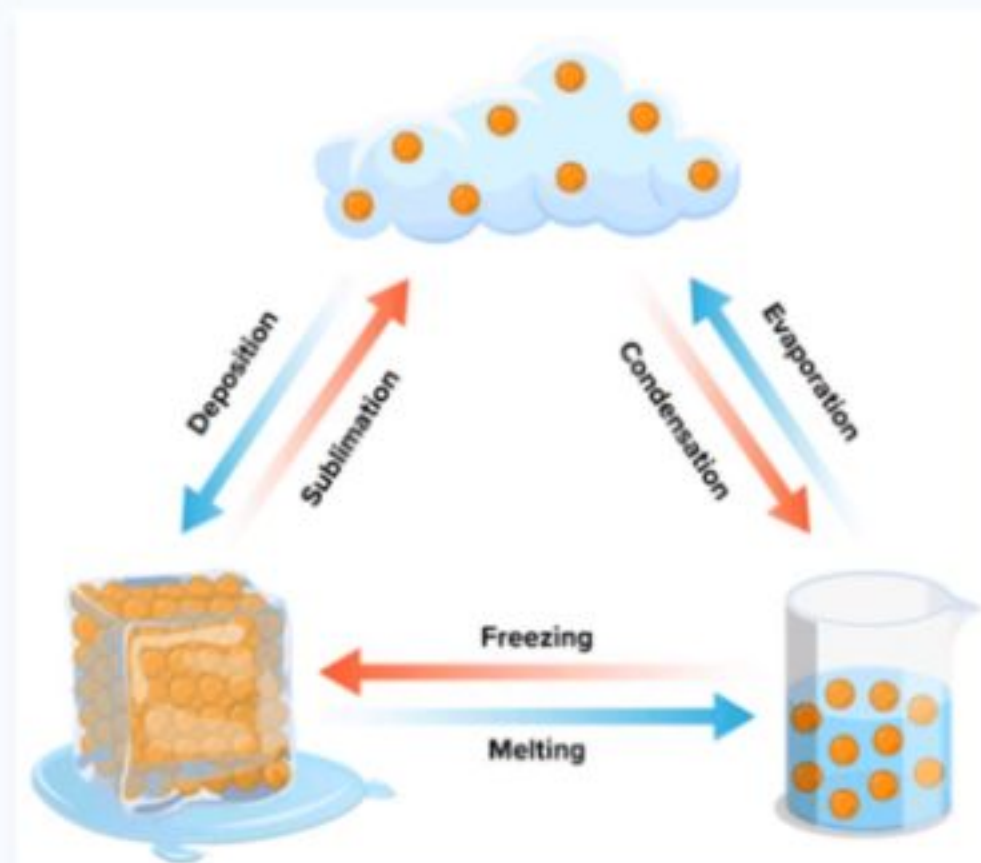
(f) Deposition

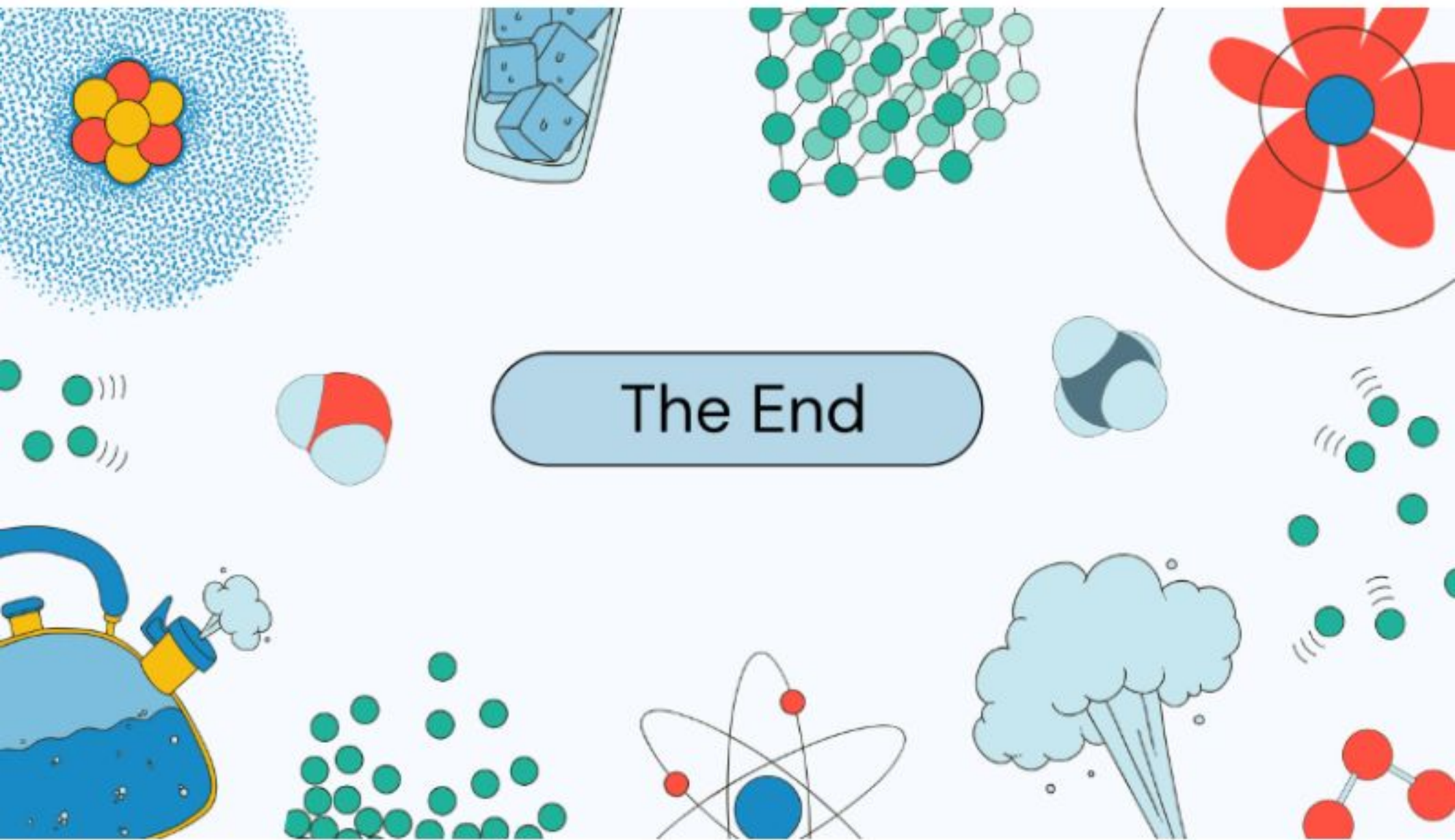
Gas → Solid directly, without becoming liquid.

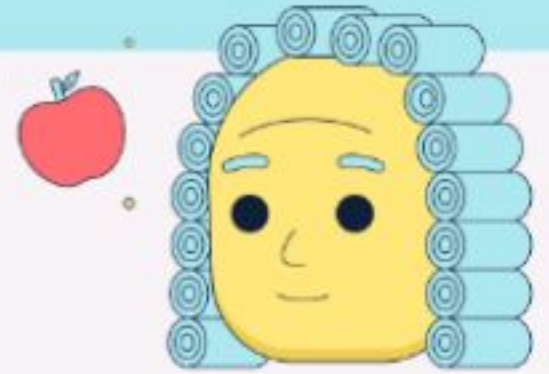
Gas particles lose energy and directly form a solid.

Example: Frost forming on a cold window from water vapour.

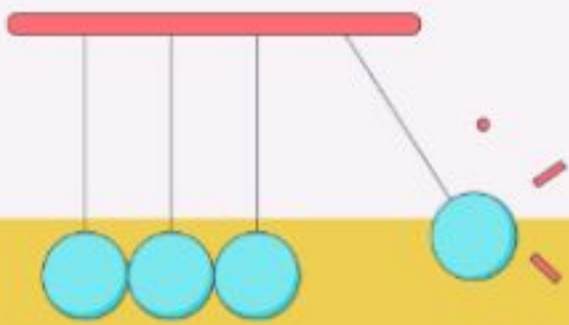
Change of state diagram



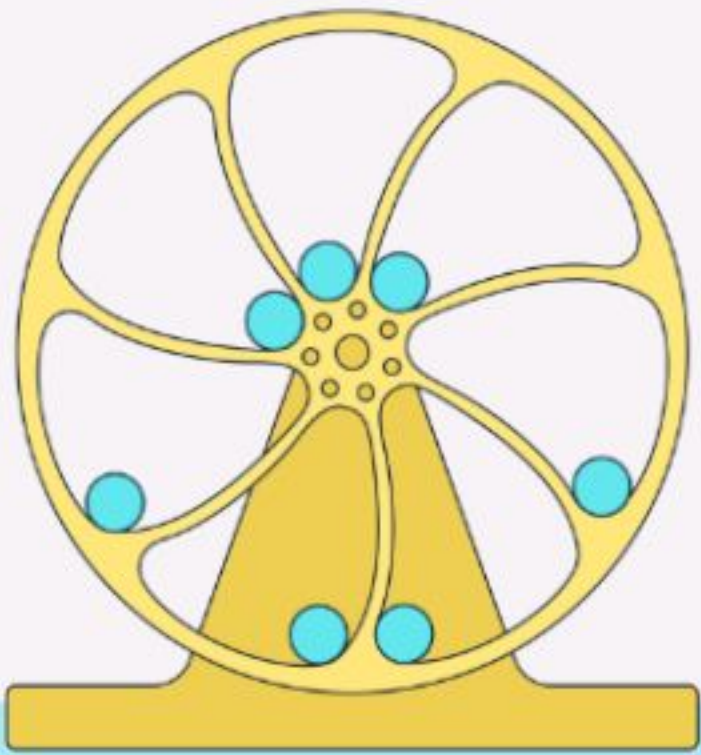




FORCES, MOTION, AND PRESSURE



Lesson Objectives



Force and Pressure
Turning effect of force(moment of force): concept, definition and calculation
Pressure:
Definition
Unit
Calculation of pressure in simple cases
Pressure exerted by liquids (Qualitative only).
Pressure exerted by gases- Atmospheric pressure (Qualitative only).

What is Force?

INTRODUCTION

Force is a push or pull acting on an object. It can change the shape, speed, direction, or state of motion of an object. A force may also stop a moving body or start a body from rest.

Effects of force

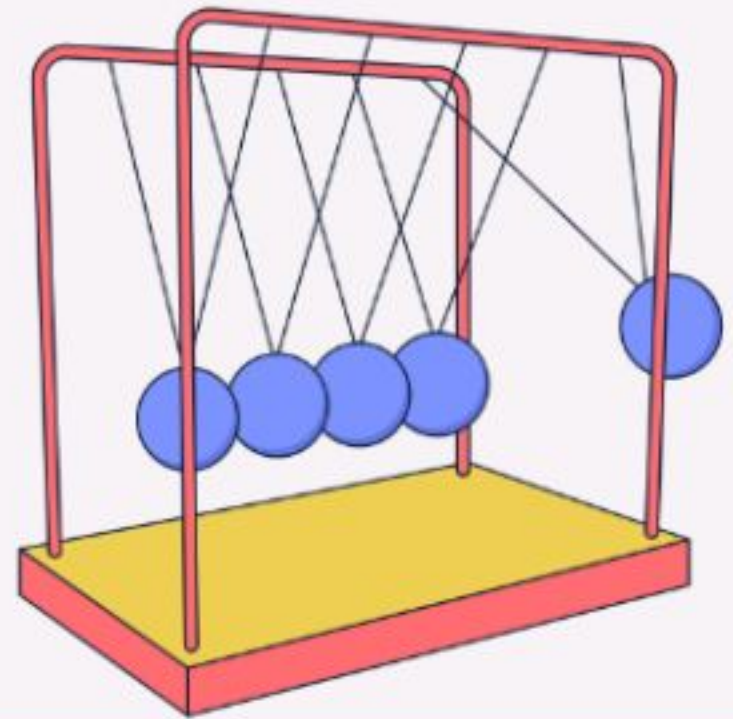
It can move a stationary object.

It can stop a moving object.

It can change the speed of an object.

It can change the direction of motion.

It can change the shape of an object.



TURNING EFFECT OF FORCE



THE TURNING EFFECT OF FORCE IS CALLED THE MOMENT OF FORCE. IT IS THE TURNING ACTION PRODUCED BY A FORCE ABOUT A FIXED POINT CALLED THE PIVOT OR FULCRUM.

- **DEFINITION:**

Moment of force is the product of the force and the perpendicular distance from the pivot to the line of action of the force.

- **FORMULA:**

Moment of force = Force $\times$ Perpendicular distance

$M = F \times d$

where: where:

- M = moment of force
- F = force
- d = perpendicular distance from pivot

The SI unit of moment of force is newton metre(Nm)

TYPES OF FORCES

CONTACT FORCES:

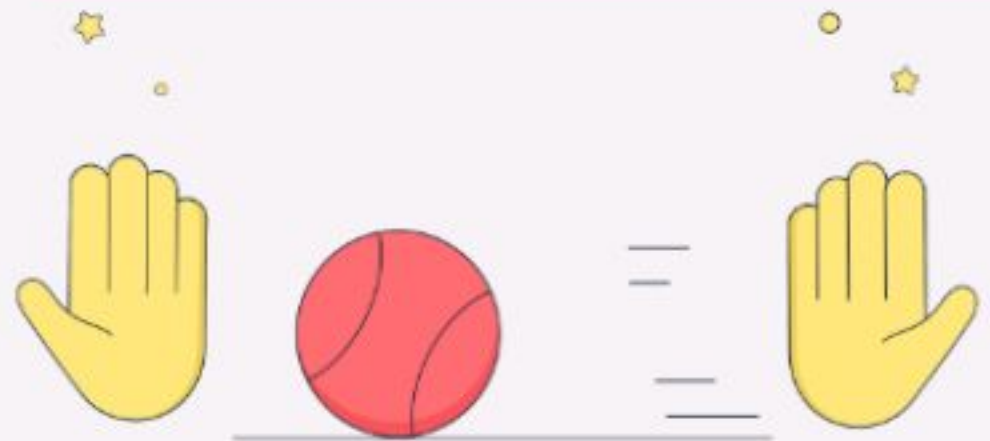
CONTACT FORCES ACT ONLY WHEN TWO OBJECTS ARE IN PHYSICAL CONTACT WITH EACH OTHER. THE CONTACT FORCES ARE AS FOLLOWS:

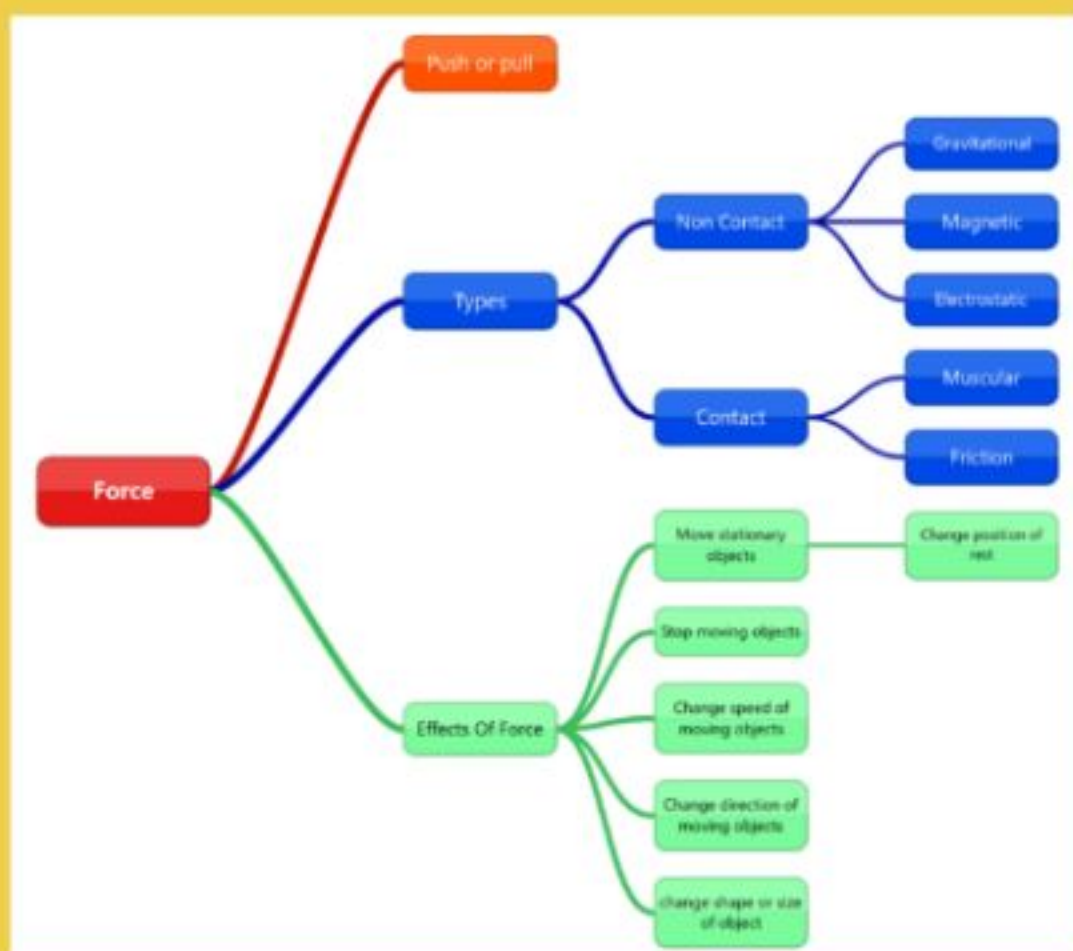
- **MUSCULAR FORCE:** THE FORCE APPLIED BY MUSCLES OF HUMANS OR ANIMALS. IT IS USED FOR PUSHING, PULLING, LIFTING, AND CARRYING OBJECTS.
- **FRICTIONAL FORCE:** THE FORCE THAT OPPOSES MOTION WHEN TWO SURFACES RUB AGAINST EACH OTHER.
- **TENSION FORCE:** THE FORCE TRANSMITTED THROUGH A STRETCHED ROPE, STRING, OR WIRE.
- **SPRING FORCE:** THE FORCE EXERTED BY A COMPRESSED OR STRETCHED SPRING.
- **AIR RESISTANCE:** THE FORCE EXERTED BY AIR ON A MOVING OBJECT, OPPOSITE TO ITS MOTION

NON-CONTACT

Non-contact forces act without physical contact between objects. They are also called field forces.

- **Gravitational force:** The force of attraction between any two bodies having mass. On Earth, it pulls objects toward the ground.
- **Magnetic force:** The force exerted by a magnet on magnetic materials or another magnet
- **Electrostatic force:** The force of attraction or repulsion between electric charges





PRESSURE

Pressure is the force acting per unit area. It tells us how much force is spread over a surface. Formula for pressure is as follows:

Pressure = Force / Area

The SI unit of pressure is pascal (Pa)

$1 \text{ Pa} = 1 \text{ N/m}^2$

For example:

If force = 20 N and area = 5 m²:

$P = 20/5$

$P = 4 \text{ Pa}$



Pressure Definition in Science

Pressure is normal force per unit area.
Units include N/m^2 or pascal (Pa).

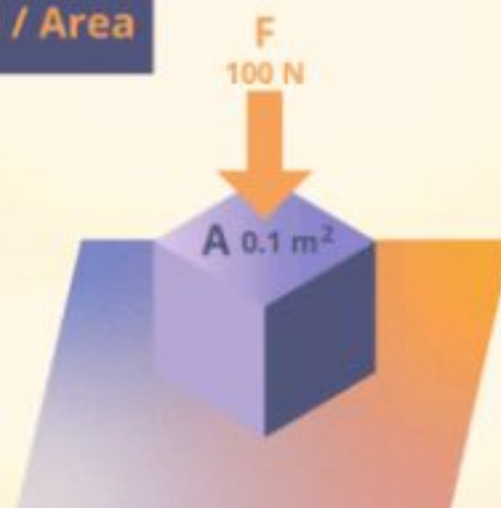
$$\text{Pressure} = \text{Force} / \text{Area}$$

$$P = F / A$$

$$P = 100 \text{ N} / 0.1 \text{ m}^2$$

$$P = 1000 \text{ N/m}^2$$

$$P = 1000 \text{ Pa}$$



sciencenotes.org

Types of pressure

1

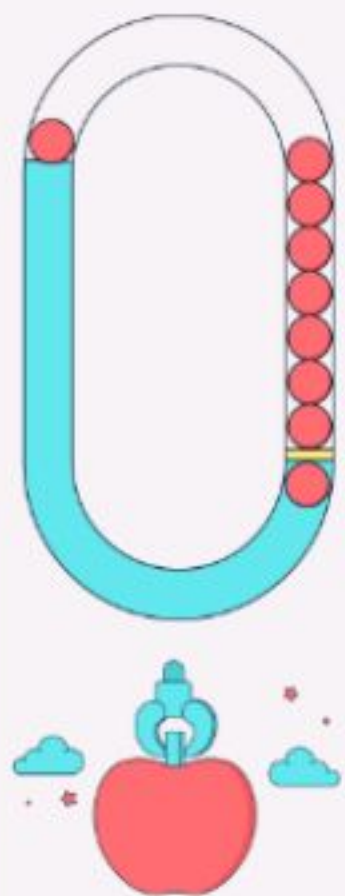
Pressure exerted by liquids
Liquids exert pressure on the walls and bottom of the container. They also exert pressure in all directions. The pressure increases as depth increases because more liquid lies above the point.

2

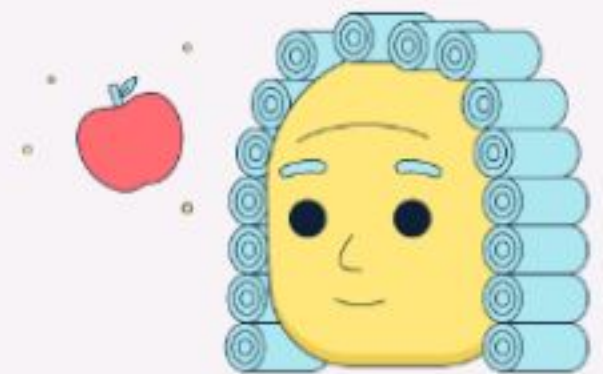
Gases also exert pressure because gas particles move constantly and collide with the walls of their container. This pressure acts in all directions.

3

Atmospheric pressure is the pressure exerted by the air of the atmosphere on everything around us. Air has weight, so it presses on the Earth and all objects on it.



PRESSURE



1) Qualitative points for liquid

- Liquids exert pressure due to their weight.
- Liquid pressure increases with depth.
- Liquid pressure acts in all directions.
- The lower parts of a container feel more pressure than upper parts.

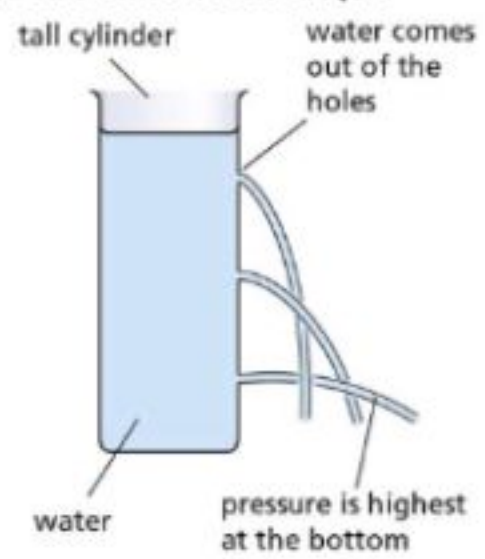
2) Qualitative points for gas

- Gas molecules keep moving randomly.
- Their collisions with the walls produce pressure.
- Gas pressure acts in all directions.
- Air inside a balloon pushes outward because of gas pressure.

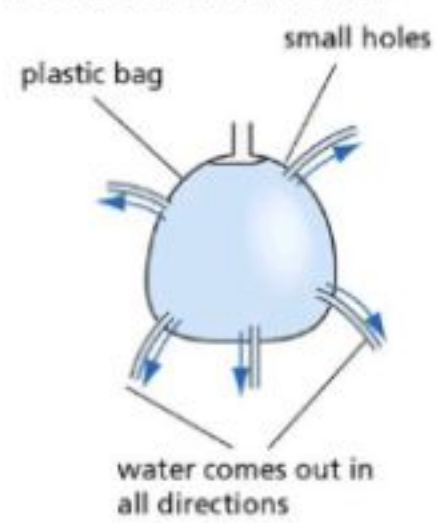
3) Qualitative points for atmospheric pressure

- Atmospheric pressure is caused by the weight of air.
- It acts in all directions.
- It decreases as height increases.
- At sea level, it is greater than at hilltops or mountains.

Pressure increases with depth



Pressure acts in all directions



RECAP

Topic	Main idea	Formula/Unit
Force	Push or pull	—
Moment of force	Turning effect of force	$M = Fd \sin \theta$
Pressure	Force per unit area	$P = F/A$, unit: Pa
Liquid pressure	Increases with depth	Qualitative
Gas pressure	Due to collisions of gas molecules	Qualitative
Atmospheric pressure	Pressure due to air	Qualitative

END